

CO1 Assessment Tool-1

- ① What is Data Science? Explain its primary objective.
- Data Science is the field of extracting useful knowledge and insights from data using scientific methods, algorithms, and statistical techniques.
- Primary objective: To analyze data and support better decision-making and predictions.
- ② Why is Data Science important in today's digital world? Mention any four reasons.
- Helps in better decision-making.
 - Improves business efficiency.
 - Enables predictive analysis.
 - Supports automation and innovation.
- ③ List any four benefits of Data Science in business and society.
- Better decision-making
 - Improved customer satisfaction
 - Fraud detection and risk management.
 - Healthcare and disease prediction.
- ④ Mention four real-world applications of Data Science.
- Healthcare
 - Banking and Finance
 - E-commerce
 - Social Media
- ⑤ What are the different facets of data? Briefly explain each.
- Structured Data: Organized in tables (e.g., databases)
 - Semi-Structured Data: Partially organized (e.g., XML, JSON)
 - Unstructured Data: No fixed format (e.g., images, videos, emails)

⑥ Differentiate between structured, semi-structured and unstructured data with suitable examples.

Data Type	Description
Structured	Stored in rows and columns
Semi-structured	Has tags but no fixed table
Unstructured	No predefined format.

⑦ What is Big Data? State its key characteristics.

Big Data refers to extremely large and complex datasets that cannot be processed using traditional tools.

5 Vs:

- Volume
- Velocity
- Variety
- Veracity
- Value

⑧ Explain the Big Data ecosystem. Name any four technologies used in it.

The Big Data ecosystem is a collection of tools used to store, process, analyze, and visualize large datasets.

Technologies:

- Hadoop
- Spark
- Hive
- HBase

⑨ What are the major phases of the Data Science process?

1. Business Understanding
2. Data Collection
3. Data preparation
4. Data Analysis
5. Model Building
6. Model Evaluation
7. Deployment
8. Communication

10) What is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of obtaining data from different storage locations.

Sources:

- Databases
- Web APIs

11) Why is data collection considered an important step in the Data Science process?

It provides accurate and relevant data for analysis, helping build reliable models and improve prediction accuracy.

12) What is data cleansing? List any four common data-cleaning techniques.

Data cleansing is the process of correcting or removing inaccurate or incomplete data.

Techniques:

- Remove duplicate records
- Handle missing values
- Correct inconsistent data.
- Remove outliers

13) What is data integration? Why is it necessary in Data Science?

Data Integration is the process of combining data from multiple sources into a single dataset.

Importances:

- Improves data quality.
- Provides a complete view of information.
- Supports better analysis.
- Eliminates redundancy.

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Explain data transformation with suitable examples.
Data transformation is the process of converting data into a suitable format for analysis.

Examples:

- Converting text to numbers.
- Scaling values between 0 and 1.
- Changing data format.

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What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining data to discover useful information and patterns.

Techniques:

- Descriptive Analysis
- Diagnostic Analysis
- Predictive Analysis
- Prescriptive Analysis

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